

CSE 3300/5299 Homework 2 (Due on 03/10/2025 Midnight)

Question 1 (20 points)

a. Suppose you have the following 2 bytes: 01011100 and 01100101. What is the 1s complement of the sum of these 2 bytes? (5 pts)

$$\text{sum: } 01011100 + 01100101 = 11000001$$

$$\text{1s complement: } 00111110$$

b. Suppose you have the following 2 bytes: 11011010 and 01100101. What is the 1s complement of the sum of these 2 bytes? (5 pts)

$$\text{sum: } 11011010 + 01100101 = 10011111 = 01000000$$

$$\text{1s complement: } 10111111$$

c. For the bytes in part (a), give an example where one bit is flipped in each of the 2 bytes and yet the 1s complement doesn't change. (5 pts)

if you flip the last bit in both bytes (making them 01011101 and 01100100), the sum is still 11000001 and the 1s complement would still be 00111110

d. UDP and TCP use 1s complement for their checksums. Suppose you have the following bitstream: 0101 0011 0110 0110 0111 0100. What is the checksum? (It is 1s complement of the sum of this bitstream. Note that although UDP and TCP use 16-bit words in computing the checksum, for this problem you are being asked to consider 8-bit sums.) Show all work. (5 pts)

$$\begin{array}{r} \begin{array}{r} \overset{1}{0}101\overset{1}{0}0\overset{1}{0}11 \\ + 01100110 \\ \hline 1'0'1'11001 \\ + 01110100 \\ \hline 100101101 \end{array} \quad \begin{array}{l} \text{carryover:} \\ 100101101 + 1 = 00101110 \\ \text{1s complement:} \\ 11010001 \end{array} \end{array}$$

Question 2 (20 pts)

Two hosts X and Y are communicating over a TCP connection, all bytes up through byte 163 being sent by Host X and received by Host Y. Now assume that Host X sends three segments to Host B back-to-back. The first segment contains 40 bytes, the second segment contains 50 bytes, and the third segment contains 80 bytes.

In the first segment, the sequence number is 164, the source port number is 105, and the destination port number is 80. Host Y sends an acknowledgment when it receives a segment from Host X.

a. In the third segment sent from Host X to Y, what are the sequence number, source port number, and destination port number? (3 pts)

$$\text{sequence number: } 164 + 40 + 50 = 254$$

$$\text{source port number: } 105$$

$$\text{destination port number: } 80$$

b. If the first segment arrives before the second segment, in the acknowledgment of the first arriving segment, what is the acknowledgment number, the source port number, and the destination port number? (5 pts)

sequence number: $164 + 40 = 204$

source port number: 80

destination port number: 105

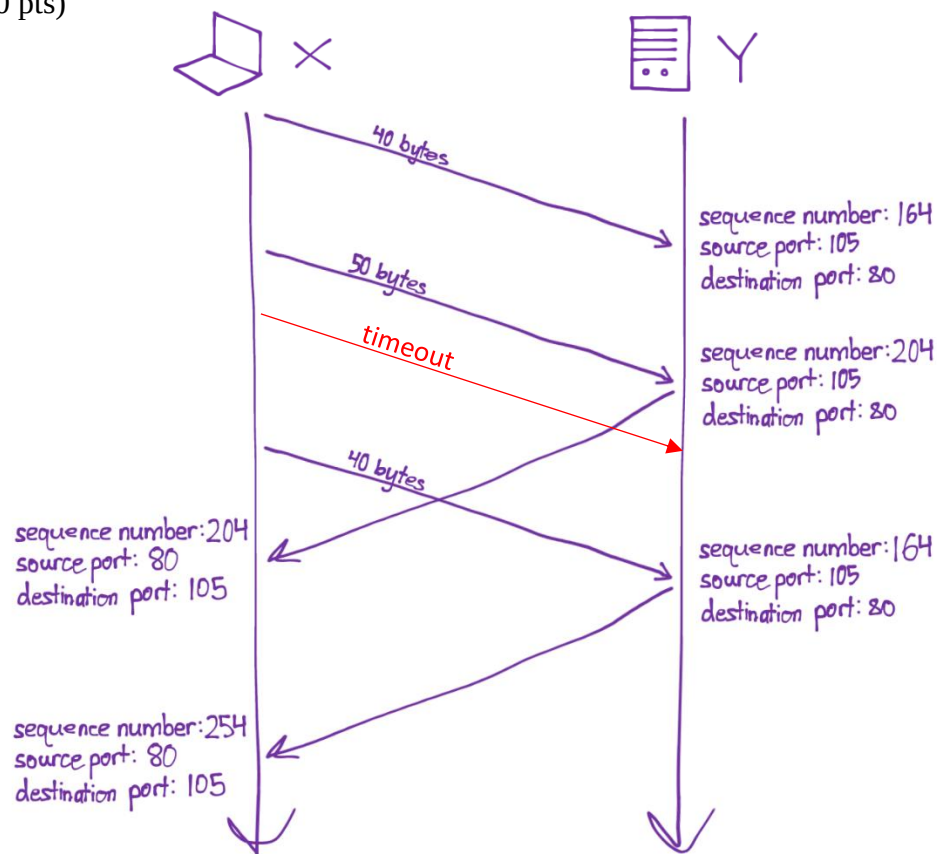
c. If the second segment arrives before the first segment, in the acknowledgment of the first arriving segment, what is the acknowledgment number? (2 pts)

sequence number: 164 (still waiting for first set of bits)

source port number: 80

destination port number: 105

d. Suppose the two segments sent by X arrive in order at Y. The first acknowledgment is lost, and the second acknowledgment arrives after the first timeout interval. Draw a timing diagram, showing these segments and all other segments and acknowledgments sent. (Assume there is no additional packet loss.) For each segment in your figure, provide the sequence number and the number of bytes of data; for each acknowledgment that you add, provide the acknowledgment number. (10 pts)



Question 3 (5 pts)

In design of the reliable data transfer protocol, what mechanism is used to handle the case that the receiver may receive a segment with errors?

The checksum mechanism is used to detect errors and the ACK or NAK protocol is used to alert the sender that they received a segment with errors.

Question 4 (5 pts)

In protocol rdt3.0 (unreliable channel can cause bit errors and packet loss), the ACK packets flowing from the receiver to the sender do not have sequence numbers (although they do have an ACK field that contains the sequence number of the packet they are acknowledging). Why is it that our ACK packets do not require sequence numbers?

They do not require sequence numbers because ACKs acknowledge individual data packets, so the packets themselves don't need sequence numbers when the ACK can serve as an identification of what packet is being acknowledged, or the timeout can serve as a flag that the packet wasn't sent.

Question 5 (10 pts)

Suppose that the five measured *SampleRTT* values are 106 ms, 120 ms, 140 ms, 90 ms, and 115 ms. Compute the *EstimatedRTT* after each of these SampleRTT values is obtained, using a value of and assuming that the value of *EstimatedRTT* was 100 ms just before the first of these five samples were obtained. Compute also the *DevRTT* after each sample is obtained, assuming a value of and assuming the value of *DevRTT* was 5 ms just before the first of these five samples was obtained. Last, compute the TCP *TimeoutInterval* after each of these samples is obtained. (beta = 0.25, alpha = 0.125)

$$\begin{aligned} \text{devRTT} &= (1-\beta) \cdot \text{devRTT}_{\text{prev}} + \beta \cdot |\text{sampleRTT} - \text{estRTT}_{\text{prev}}| \\ \text{estRTT} &= (1-\alpha) \cdot \text{estRTT}_{\text{prev}} + \alpha \cdot \text{sampleRTT} \\ \text{TimeoutInterval} &= 4 \cdot \text{devRTT} + \text{estRTT} \end{aligned}$$

sample 1: 106 ms

$$\begin{aligned} \text{devRTT} &= (1-0.25) \cdot 5 + 0.25 \cdot |106 - 100| = 5.25 \text{ ms} \\ \text{estRTT} &= (1-0.125) \cdot 100 + 0.125 \cdot 106 = 100.75 \text{ ms} \\ \text{Timeout} &= 4 \cdot 5.25 + 100.75 = 121.75 \text{ ms} \end{aligned}$$

sample 2: 120 ms

$$\begin{aligned} \text{devRTT} &= (1-0.25) \cdot 5.25 + 0.25 \cdot |120 - 100.75| = 8.75 \text{ ms} \\ \text{estRTT} &= (1-0.125) \cdot 100.75 + 0.125 \cdot 120 = 103.16 \text{ ms} \\ \text{Timeout} &= 4 \cdot 8.75 + 103.16 = 138.16 \text{ ms} \end{aligned}$$

sample 3: 140 ms

$$\begin{aligned} \text{devRTT} &= (1-0.25) \cdot 8.75 + 0.25 \cdot |140 - 103.1625| = 15.77 \text{ ms} \\ \text{estRTT} &= (1-0.125) \cdot 103.1625 + 0.125 \cdot 140 = 107.76 \text{ ms} \\ \text{Timeout} &= 4 \cdot 15.7734375 + 107.7617188 = 170.86 \text{ ms} \end{aligned}$$

sample 4: 90 ms

$$\begin{aligned} \text{devRTT} &= (1-0.25) \cdot 15.7734375 + 0.25 \cdot |90 - 107.7617188| = 16.27 \text{ ms} \\ \text{estRTT} &= (1-0.125) \cdot 107.7617188 + 0.125 \cdot 90 = 105.54 \text{ ms} \\ \text{Timeout} &= 4 \cdot 16.27050783 + 105.541504 = 170.62 \text{ ms} \end{aligned}$$

sample 5: 115 ms

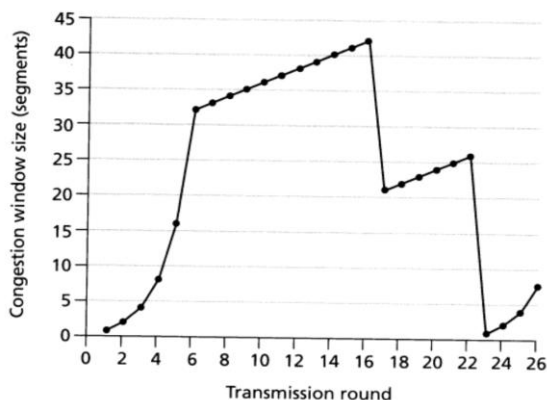
$$\begin{aligned} \text{devRTT} &= (1-0.25) \cdot 16.27050783 + 0.25 \cdot |115 - 105.541504| = 14.57 \text{ ms} \\ \text{estRTT} &= (1-0.125) \cdot 105.541504 + 0.125 \cdot 115 = 106.72 \text{ ms} \\ \text{Timeout} &= 4 \cdot 14.56750487 + 106.723816 = 164.99 \text{ ms} \end{aligned}$$

Question 6 (5 pts)

We saw that TCP waits until it has received three duplicate ACKs before performing a fast retransmit. Why do you think the TCP designers chose not to perform a fast retransmit after the first duplicate ACK for a segment is received?

They chose to wait for three duplicate ACKs because sometimes packets can be received out of order, so a single duplicate doesn't necessarily mean packet loss, it could just mean out of order. Receiving three duplicates means there's a much higher probability of packet loss since it being *that* shuffled is less likely.

Question 7 (35 pts) Consider the figure below.



Assuming TCP Reno is the protocol experiencing the behavior shown, answer the following questions. In all cases, you should provide a short discussion justifying your answer.

- Identify the intervals of time when TCP slow start is operating. (5 pts)
occurs when it increases exponentially, which is at 1-6 and 23-26
- Identify the intervals of time when TCP congestion avoidance is operating. (5 pts)
occurs when it increases linearly, which is at 6-16 and 17-22
- After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout? (5 pts)
It dropped from 42 segments to 21 segments, which a TCP Reno protocol does when a triple duplicate ACK is detected.
- After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout? (5 pts)
It dropped from 26 segments to 1 segment, which a TCP Reno protocol does when a timeout is detected.
- What is the initial value of ssthresh at the first transmission round? (5 pts)
it was 1 segment at round 1
- What is the value of ssthresh at the 20th transmission round? (5 pts)
it was 23 segments at round 20
- What is the value of ssthresh at the 24th transmission round? (5 pts)
it was 2 segments at round 24